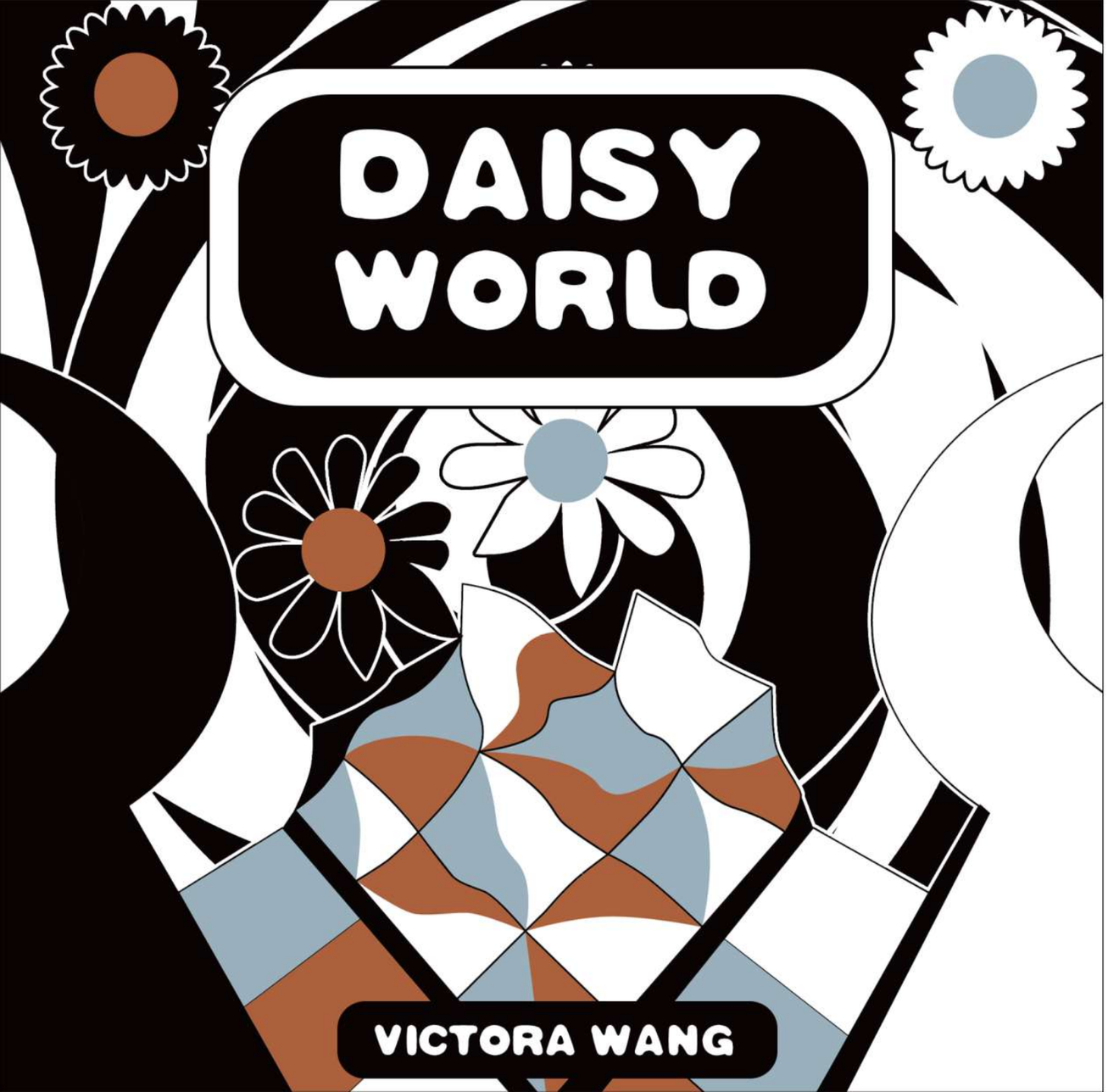


A BOARD GAME

DAISY WORLD

Designed by VICTORIA WANG

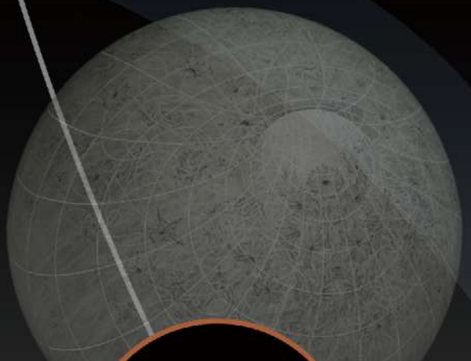
In this world, two mysterious plants hold the power to shape the climate's temperature. Choose one to cultivate and watch as the climate shifts. Will the changing world favor your growth, or will it test your strategy?

Adapt, expand, and guide your plant toward its perfect environment—only the most skillful gardener will thrive!

For the digital version, please visit: https://victoria-wang.itch.io/daisyworld
For the introduction video, please visit: https://youtu.be/4n6pi1tc-Ns

INSPIRATION

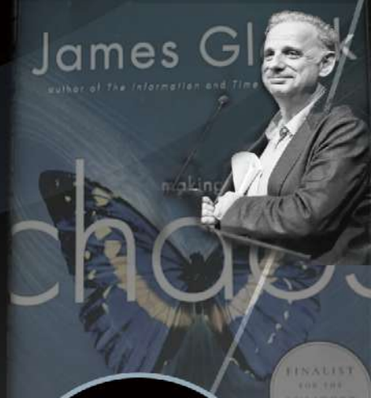
The Daisyworld model is a simplified representation of an ecological system created by James Lovelock and Andrew Watson as an illustration of **the Gaia hypothesis**, which posits that life interacts with the physical environment to create a self-regulating, stable climate conducive to sustaining life. In the model, **a hypothetical planet's temperature** is regulated by **the albedo effect** of two types of **daisies**—black and white—that either absorb or reflect heat, demonstrating how life-forms can collectively maintain equilibrium within an ecosystem.



ARTWORKS OF CAROL PRUSA

Carol Prusa creates a new vision of the powers of the universe in each artwork she makes. Inspired by cosmology and all of the **natural sciences**, Prusa creatively explores these practices, arriving at pictorially stunning re-interpretations of their theories.

Chaos theory is one of the important scientific knowledge of our time, heralding a new wave of human thought. The science writer Gleick, with his profound scientific literacy, elucidates the mysteries of chaos theory in an accessible manner. The book **CHAOS: MAKING A NEW SCIENCE** is Gleick's breakout work and a classic in the field of popular science on chaos theory.



A BOOK BY JAMES GLEICK



The Mandelbrot set is a geometric figure that was once called **the fingerprint of God**. It was discovered by Professor Mandelbrot in the seventies of the twentieth century.

MANDELBROT SET



GAIA HYPOTHESIS AND DAISY WORLD



ARTWORKS OF DANIEL OST

Daniel Ost is known worldwide as a leading floral house that brings art and nature together. With an unwavering commitment to creativity and craftsmanship, they design **floral masterpieces** that evoke emotions and tell stories.

SET UP

① PLANTING BOARD
The planting board representing the world of daisies is divided into 16 square areas, unfolded, and placed with the colored side facing up.

② TEMPERATURE PIECE
Place the temperature pieces on the black dots along the edge of the board.It represents the current temperature in the Daisy World.

③ CARDS
Shuffle the cards and place them face down. Each person will be given 1 card at the start, with each side having a maximum of 5 cards. Each side can draw up to 3 cards per turn.

④ WHITE DAISY PIECES
White daisies thrive in high temperatures, but planting them will cause the temperature to decrease. Each time a white daisy is planted/removed, the temperature decreases/increases by 2 ℃ .

⑤ BLACK DAISY PIECES
Black daisies thrive in low temperatures, but planting them will cause the temperature to increase. Each time a black daisy is planted/removed, the temperature increases/decreases by 2 ℃ .

⑥ ACTION POINT TOKENS
Tokens used to count action points. Each turn, both sides receive 2 action points. Planting daisies, removing daisies, drawing cards, and playing cards each consume 1 action point. Each card discarded earns 1 action point.

⑦ CARD SLOTS
The numbers at the bottom of the board are used to record the number of turns that the cards are effective.

⑧ SQUARE AREAS
The colors of the square areas symbolize the climate of the areas. Areas with blue are suitable for the growth of black daisies, areas with orange are suitable for the growth of white daisies, and areas with both colors are suitable for the growth of both types of daisies. Daisies of the same type in the same square can remove each other. When a square contains 3 daisies of the same type, it is considered fully occupied, and no new daisies can be planted on that area.

⑨ TEMPERATURE SCALE
The temperature scale records the temperature of the Daisy World, with each mark representing 1° C. The orange direction indicates rising temperatures, while the blue direction indicates falling temperatures. The upper edge marks the neutral temperature range; when the temperature token is here, both types of daisies can be planted. The left edge/right edge represents high/low temperature zones, where black daisies/white daisies cannot be planted.



CONTENT

PIECES ×151



Black Daisies ×50 · White Daisies ×50 · Temperature Piece ×1 · Action Point Tokens ×50

CARDS ×42

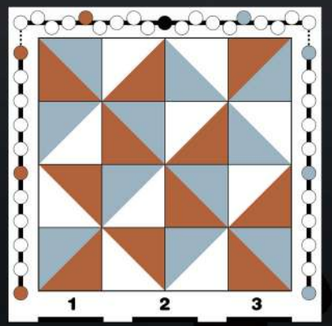


Drought\Frost ×10 · Warm\Cold Period ×5 · Expansion ×5 · Hurricane ×5



El Niño ×5 · Climate Inertia ×5 · Climate Resilient ×5 · Explanatory Cards ×2

CHESSBOARD ×1



SYSTEM DESIGN

CARD SYSTEM

At the start of the game, each player will receive a card. After that, **drawing and playing** cards will **cost action points**, while **discarding** cards can **gain action points.**



INFLUENCE TEMPERATURE

Some cards can directly **raise or lower the temperature**, and there is a card that can **remove the temperature effects** generated by the opponent in the previous turn.



INFLUENCE DAISIES

There is a card that can **remove daisies** without generating a temperature effect, and some cards can **amplify or reverse the temperature effects** produced during planting.

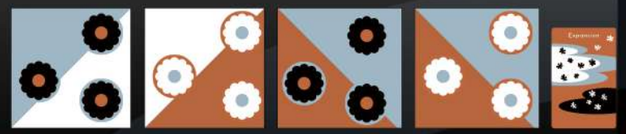


DISCARD

Each **card discarded** can get **1 action point**, drawing and playing cards consume **1 action point**, each person can draw a card up to **3 times** per turn, and at the end of the round, there shall be no more than **5 cards** in hand.

INCREASE ACTION POINTS

There is a card that can be used with daisy planting to gain **extra action points**. If you play the card Expansion, you will gain **1 more action point** for every **1 suitable area** your daisies **fully occupied** (the orange areas corresponding to white daisies while the blue areas corresponding to black daisies).



ACTION POINTS SYSTEM

At the start of each round, both sides will receive **2 action points**, and any unused action points from the previous round can **be carried over** to the next round.

VICTORY CONDITIONS 1

When the temperature piece reaches **the orange end**, **the white daisies** wins; When it reach **the blue end**, **the black daisies** wins.

VICTORY CONDITIONS 2

When the temperature piece is on the **left edge** and there are **no black daisies** on the field, **the white daisies** win; When the temperature piece is on the **right side** and there are **no white daisies** on the field, **the black daisies** win.

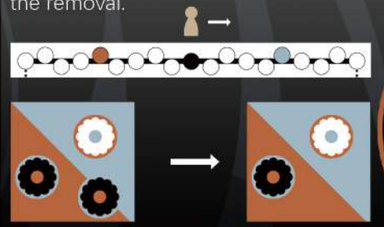
PLANT REMOVE

Planting or removing 1 daisy consumes **1 action point**. For each **white daisy planted**, the temperature **decreases by 2 ℃** , while for black daisies, it's the opposite. Generally, you can remove any of **your own daisies**. Each **white daisy** removed **increases** the temperature by 2 ℃ , while black daisies have the opposite effect.



INTERSPECIFIC COMPETITION

When your daisy and the opponent's daisy are in the **same square area**, you can spend **1 action point** to remove **1 of the opponent's daisies** in that area and calculate the temperature effect caused by the removal.

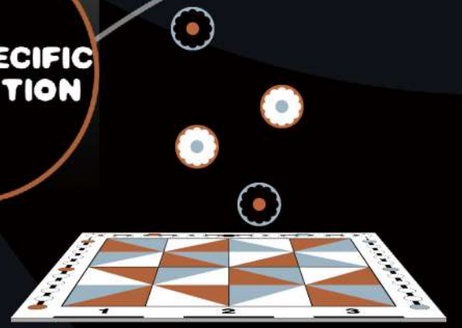


INCREASE ACTION POINTS

When a square contains 3 daisies of the same type, it is considered **fully occupied**. At this time, when combined with certain cards, you can gain a **temperature boost**. When the fully controlled area is a **suitable area**, the boost effect will expand, and you can also gain **action points** by using **Expansion**.

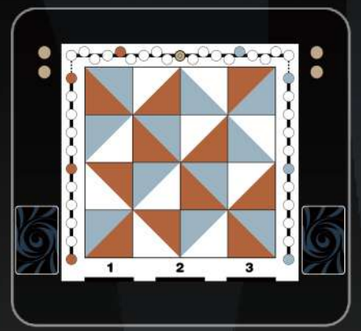
PIECES SYSTEM

The board is divided into 16 square areas, and each area can hold up to **3 daisies**. **Planting and removing** daisies both **consume action points**. For **white daisies planted**, the temperature **decreases**, while for black daisies, it's the opposite.

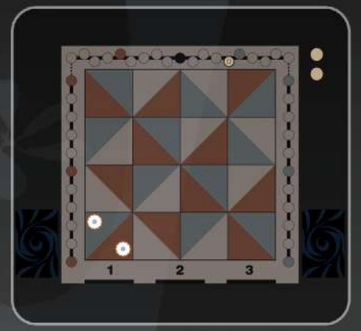


GAME PROCESS

Round 1



Before the first round begins, both sides decide who goes first by playing rock-paper-scissors. White Daisy goes first. After the first round begins, both sides receive one card and two action points, and the temperature chess piece is in a neutral position.



The white side chooses to plant 2 white daisies, and the temperature piece moves 4℃ toward low temperature.



The black side chooses to plant 1 black daisy, and the temperature piece moves 2℃ toward high temperature. Then, the black plays a card Drought/Frost, and the temperature piece moves 5℃ toward low temperature.

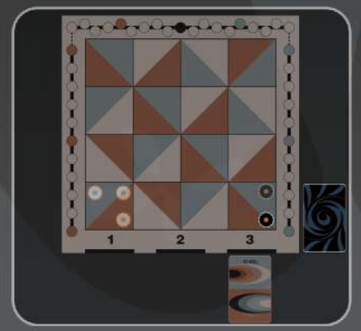
Round 2



The white side discards a card Expansion and gains 1 action point.



The white side draws a card and plays a card El Niño, and has 1 action point remaining. Then, the white side plants 1 white daisy, and due to the El Niño effect, the temperature piece moves 2℃ toward high temperature.



The black side draws 1 card and plants 1 black daisy. Due to the El Niño effect, the temperature piece moves 2 ℃ toward low temperature.

Round N



The white side discards 3 cards and gains 3 action points.



The white side removes 1 black daisy and plants 1 white daisy, and due to the El Niño effect, the temperature piece moves 4 ℃ toward high temperature.



The white side plays a Dought/Frost card, causing the temperature to rise by 5 ℃. After that, the white side plays a Hurricane card, removing all daisies from the 2×2 square in the lower right corner. With the temperature in the high range and no black daisies on the field, the White

ESTABLISH THE BASIC SETTINGS AND RULES

1. Game Board: 6x6 grid, each square can plant 3 daisies.

2. Pieces and Temperature System: The global temperature determines the survival conditions for both daisies, ranging from -10 ° C to +50 ° C, with an initial temperature of 20 ° C (neutral state). Temperature changes are triggered by planting/removing either black or white daisies. Each planted black/white daisy +/- 2 ° C, while each removed daisy -/+1° C.

3. Cards: Each turn, you may draw one card. Cards are divided into numerical cards and rule cards.
Numerical cards: Drought\Frost×5, Warm\Cold Period×5.
Rule cards: Climate Inertia×3, El Niño×3, Paris Agreement×3, Climate Resilience×3.

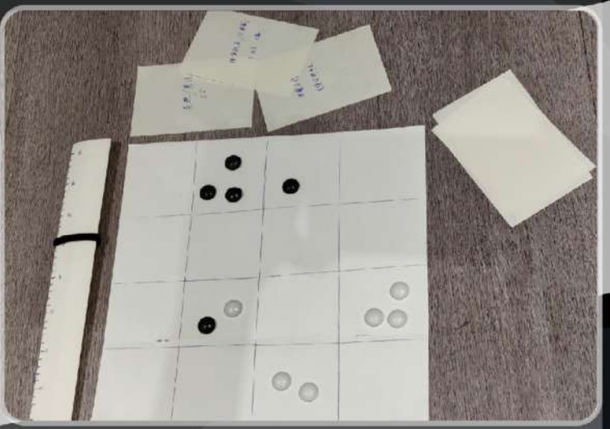
DETERMINE THE VICTORY CONDITIONS

1. After filling the board, calculate the survival control area. The side with the higher score wins. If scores are tied, the side closer to the temperature wins.

2. Black Daisy: Lower the temperature below 0° C and maintain it for 3 turns.
White Daisy: Raise the temperature above 40° C and maintain it for 3 turns.

3. One side eliminates all of the other side's daisies.

STEP 1



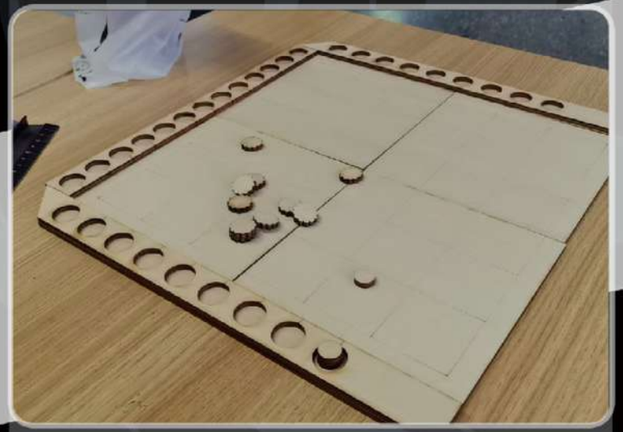
SYSTEM MODIFICATION

1. Action Point System Added: Each turn, both players have 1 action point. Planting/removing/moving daisies (1 point), destroying an opponent's daisy on a tile (1 point), using a card (1 point).

2. The Card-Drawing Mechanism Changed: Both players start with one card in the first round. Subsequent rounds require spending action points to draw cards, with a maximum hand size of five cards.

3. The Occupation Rule Changed: A complete occupation is achieved when three daisies of the same color occupy a single square. If daisies of different colors coexist in the same square, you may spend 1 action point to remove the opposing daisy and calculate the temperature benefit.

STEP 2



THE VICTORY CONDITIONS CHANGED

1.When the temperature cursor reaches the orange terminus, the white daisy wins; When you reach the blue end, the black daisy wins.

2.When the temperature verifier is on the left edge and there are no black daisies on the field, the white daisies win; When the temperature rover is on the right side and there are no white daisies on the field, the black daisies win.

NUMERICAL ADJUSTMENT

1. Action Points: Each turn, both players have 2 action points. Discarding 1 card grants 1 point.

2. Temperature System: The global temperature ranges from 0 ° C to 40 ° C, with an initial temperature of 20 ° C (neutral state). Each removed black/white daisy ±2° C.

3. Cards:
Modification: Warm\Cold Period×5, Climate Resilience×5, Climate Inertia×5, El Niño×5.
Addition: Hurricane×5, Expansion×5.
Deletion: Paris Agreement.

STEP 3

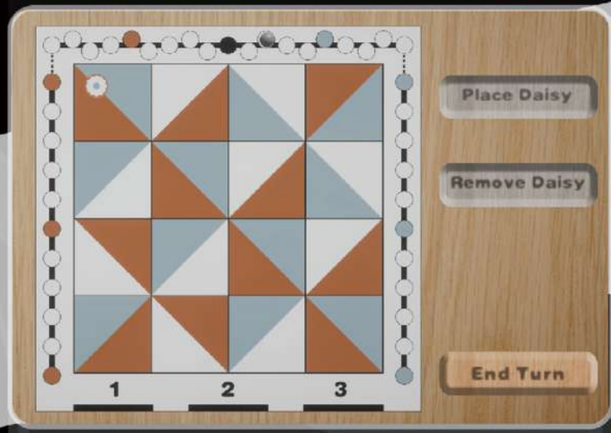


MATERIAL SELECTION

Game Board: Foldable tabletop game board.
Black and White Daisy Pieces: 2cm × 8mm wooden pieces + waterproof decals.
Neutral Pieces: Wooden carved humanoid pieces.
Cards: 62mm × 100mm blue-core playing cards.
Action Point Tokens: 2cm × 4mm round wooden discs.

TESTING

STEP 4



DEVELOPMENT OF THE DIGITAL VERSION

Implement the chessboard system and the hand card system, and enable offline human-computer battle functionality. The remaining features are still under development.

FINAL SHOW

